

Kartoteka Mieszkańca - Requirements

1. Overview

The Citizen Registry (Kartoteka Mieszkańca) is a system for storing and managing citizen data. It allows authorized users to create, view, update, and search citizen records containing personal information, contact details, identity documents, and registration history.

2. User Stories

2.1 Citizen Management

- **US-01** As a clerk, I want to register a new citizen so that their data is stored in the system.
- **US-02** As a clerk, I want to view a citizen's full record so that I can access their information.
- **US-03** As a clerk, I want to update a citizen's data so that the records remain current.
- **US-04** As a clerk, I want to delete a citizen's record so that outdated entries can be removed.
- **US-05** As a clerk, I want to search for citizens by name, PESEL, or address so that I can quickly find a specific record.

2.2 Identity Documents

- **US-06** As a clerk, I want to add identity documents to a citizen's record so that their IDs are tracked.
- **US-07** As a clerk, I want to view a history of a citizen's identity documents so that I can see past and current IDs.

2.3 Registration History

- **US-08** As a clerk, I want to record address registration changes so that a citizen's residence history is maintained.
- **US-09** As a clerk, I want to view a citizen's registration history so that I can see their past addresses.

2.4 Authentication & Authorization

- **US-10** As an administrator, I want to manage user accounts so that only authorized personnel can access the system.
- **US-11** As a user, I want to log in to the system so that I can perform my duties.

3. Functional Requirements

3.1 Citizen Record - Basic Personal Data

ID	Requirement
FR-001	The system shall store citizen's first name (required).
FR-002	The system shall store citizen's last name (required).
FR-003	The system shall store citizen's PESEL number (required, unique, 11 digits).
FR-004	The system shall validate PESEL using the checksum algorithm.

FR-005	The system shall store citizen's date of birth (derived from PESEL).
FR-006	The system shall store citizen's sex (derived from PESEL).
FR-007	The system shall store citizen's registered address (street, building number, apartment number, postal code, city).

3.2 Citizen Record - Contact Information

ID	Requirement
FR-008	The system shall store citizen's phone number (optional).
FR-009	The system shall store citizen's email address (optional).
FR-010	The system shall store citizen's correspondence address if different from registered address (optional).

3.3 Identity Documents

ID	Requirement
FR-011	The system shall store identity document type (ID card, passport).
FR-012	The system shall store document series and number.
FR-013	The system shall store document issue date and expiry date.
FR-014	The system shall store the issuing authority.
FR-015	The system shall support multiple documents per citizen.
FR-016	The system shall indicate which document is currently active.

3.4 Registration History

ID	Requirement
FR-017	The system shall record each address registration change with an effective date.
FR-018	The system shall distinguish between permanent and temporary registration.
FR-019	The system shall maintain the full history of address changes.
FR-020	The system shall store the previous address when a registration changes.

3.5 Search & Lookup

ID	Requirement
FR-021	The system shall allow searching citizens by first name, last name, or both.

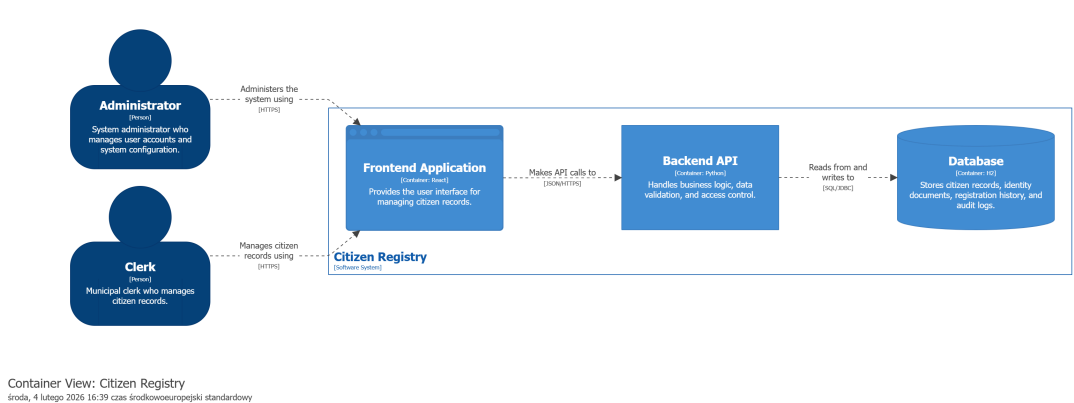
FR-022	The system shall allow searching citizens by PESEL number.
FR-023	The system shall allow searching citizens by address (city, street).
FR-024	The system shall support partial matching for name searches.
FR-025	The system shall return paginated search results.

3.6 Authentication & Authorization

ID	Requirement
FR-026	The system shall require user authentication before granting access.
FR-027	The system shall support role-based access control (administrator, clerk).
FR-028	The system shall log all data modification operations with user and timestamp.

4. Application Architecture

The application follows a classic three-tier architecture, as shown in the container diagram below.



4.1 Containers

- **Frontend Application** (React) - A single-page application running in the user's web browser. Provides the user interface for managing citizen records, searching, and viewing registration history. Communicates with the backend exclusively via HTTPS.
- **Backend API** (Python) - The server-side application that exposes a REST API. Handles all business logic including PESEL validation, role-based access control, data validation, and audit logging. Acts as the single point of access to the database, ensuring that no client can interact with the data store directly.
- **Database** (H2) - Stores all persistent data: citizen records, identity documents, registration history, user accounts, and audit logs. Accessed by the Backend API via SQL/JDBC.

4.2 Actors

- **Clerk** - A municipal clerk who uses the system to manage citizen records (create, read, update, delete, search).
- **Administrator** - A system administrator who manages user accounts and system configuration.

4.3 Communication

All communication between actors and the system occurs over HTTPS. The Frontend Application communicates with the Backend API using JSON over HTTPS. The Backend API communicates with the Database using SQL over JDBC.

5. Non-Functional Requirements

ID	Requirement
NFR-001	The system shall comply with GDPR and Polish personal data protection laws.
NFR-002	All data in transit shall be encrypted using TLS.
NFR-003	All data at rest shall be encrypted.
NFR-004	The system shall maintain an audit log of all access to citizen records.
NFR-005	The system shall be available as a web application.
NFR-006	The system shall support concurrent access by multiple users.